INCH-POUND
MIL-PRF-49450/9(CR)
10 March 2000

PERFORMANCE SPECIFICATION SHEET

BATTERY, RECHARGEABLE, NICKEL-CADMIUM, VENTED, AIRCRAFT, BB-432B/A

This specification is approved for use by the USACECOM, Department of Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-49450(CR)

Voltage. The voltage of the battery shall be 24 Vdc (nominal)

Capacity. The battery capacity shall be 10 ampere hours.

Weight. The weight of the battery shall be 34 pounds (Max)

Connector (J1). The main power connector shall be quick disconnect type that meets the dimensional and interface requirements of MS3509.

Connector (J2) The connector for the electrical harness shall be a 10 pin connector that meets the interface requirements of MS3474W12-10S.

Cells. The battery shall consist of 19 BB-599A/A cells in series.

AMSC N/A FSC 6140
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DRAWING NOTES:

1. All dimensions are in inches, unless specified otherwise.

2. Vent Pipes The battery shall have two vent pipes shown in figures 1, 2 and 3. The vent pipes shall be made of the same material as the battery case.

3. Holding strap The holding strap shown in figure 1 and figure 3 shall be located on the centerline of the battery.

4. Main power receptacle (J1). The battery shall utilize a two-pin receptacle conforming to the interface requirements of MS3509. The J1 receptacle shall be centered as shown in figure 1.

5. Sensor receptacle. (J2). The J2 receptacle shall be centered as shown in figure 1.

6. Battery Container: The body and cover shall be made of the same alkali-resistant material. The dimensions of the container and the dimensions and locations of the receptacles mounting holes and hold down strap shall conform to figure 1 and figure 2.

7. Harness assembly. The harness assembly is built into the battery and is not a separate assembly. Wiring and wire color shall be in accordance with figure 4. The harness shall consist of the following:

a. One auxiliary receptacle (J2), conforming to the interface requirements of an MS3474W12-10S receptacle, to provide connection to battery sensing and control points. The receptacle shall be mounted into a “D” hole in the battery case. The hole shall be 0.923” Dia and 0.824” across the flat. The flat surface shall face the top of the case. The location of J2 is shown on figure 1. The back of the connector shall be potted to prevent creepage of electrolyte.

b. One temperature switch mounted on a removable link. The temperature switch shall have normally open contacts, which close at 60$\pm$2.8°C and reopen at 48.8$\pm$2.8°C. The temperature switch shall be certified to withstand a minimum of 10,000 operating cycles at 30VDC and 1.0 ampere.

c. One temperature sensor (thermistor) shall be mounted on a removable link. Potting shall be employed to isolate the temperature sensor from any unconnected battery voltage.

d. One center tap voltage lead, having a terminal lug, shall be connected to the positive terminal of cell 10 (referenced from battery negative) and shall be connected to pin “F” of connector J2.

e. All wire used for electrical connections in the harness assembly shall be #20AWG, insulated and capable of withstanding the temperature and exposure to battery electrolyte.

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7.1 Harness temperature shock. The harness assembly shall be subject to the Thermal Shock Test per MIL-PFR-49450. During the test the following electrical test shall be done on the harness:

Step 1. Check for continuity between pin F of wiring harness connector and cell 10 (+) terminal. Resistance should be 0.1 ohm maximum .

Step 2. Check for continuity between pin A of wiring harness connector and battery positive (+) terminal. Resistance should be 0.1 ohm maximum .

Step 3. Check resistance between pin J of wiring harness connector and battery negative (-) terminal. Resistance should be 0.1 ohm maximum.

Step 4. Check for continuity of the thermal switch S1 between pins D and E of wiring hardness connector. Resistance should be 0.2 ohm maximum at temperatures below 135°F. Resistance should be infinite (open circuit) at temperatures above 140°F.

Step 5. Check thermistor, resistance at pins A and C of wiring harness connector. Resistance versus battery temperature should be as specified below.

Temperature	Acceptable Resistance Range (ohms)
50° F (10° C)	1800-400
60° F (16° C)	1350-700
70° F (21° C)	1050-300
80° F (27° C)	800-1000
90° F (32° C)	650-775
100° F (38° C)	500-625
110° F (43° C)	450-500

8 Color. The color of the container and cover except latches, hold-down hooks, other external hardware, pads, and identification markings and safety labels shall conform to FED-STD-595 within the range of blue color chips 15090 to 15193.

REQUIREMENTS:

Nomenclature:	BB-432B/A Battery, Rechargeable
Rated capacity:	10 ampere-hours
Voltage (Nominal):	24 Volts
Weight (Maximum):	34 pounds
Connector:	one 2 pin that meets the interface requirements of MS3509(J1)
Connector:	one 10 pin that meets the interface requirements of MS3474W12-10S(J2)

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Test	Discharge Rate(Amps)	Cut Off Voltage(V)	Capacity(AH) Minimum
Capacity	7	18	10
Cycle Life			
Normal Rate	10	18	9
High Rate	100	14.4	6.6
Vibration	10	18	10
Temperature Shock	10	18	10
Altitude	10	18	8.5
Humidity and Charge Retention	10	18	7.5
High Rate			
Discharge and Low Temperature	100	13	5
High Rate			
Discharge and Operating Position	100	14.4	8.5
Shock	10	18	10

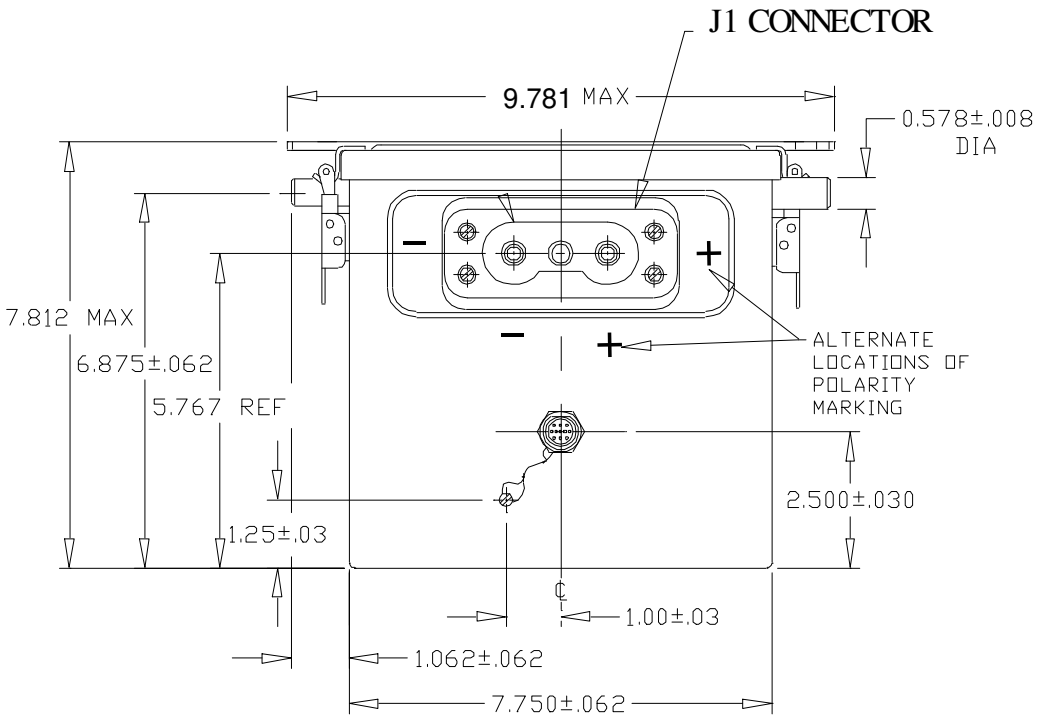


FIGURE 1: BATTERY ASSEMBLY BB-432B/A FRONT VIEW

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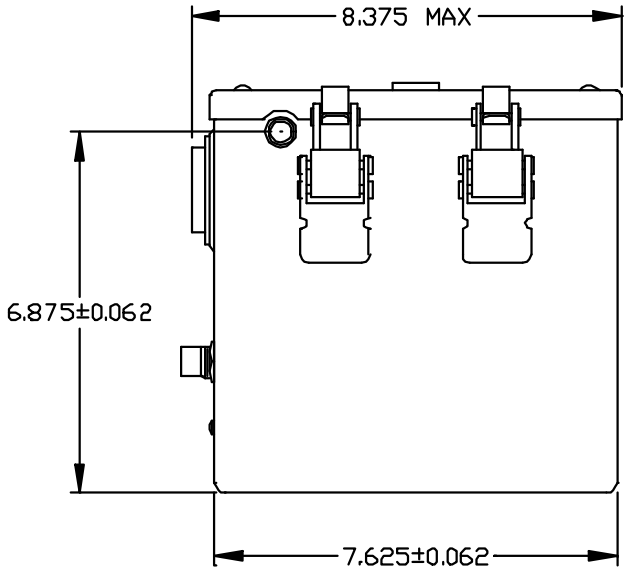


FIGURE 2: BATTERY ASSEMBLY BB-432B/A SIDE VIEW

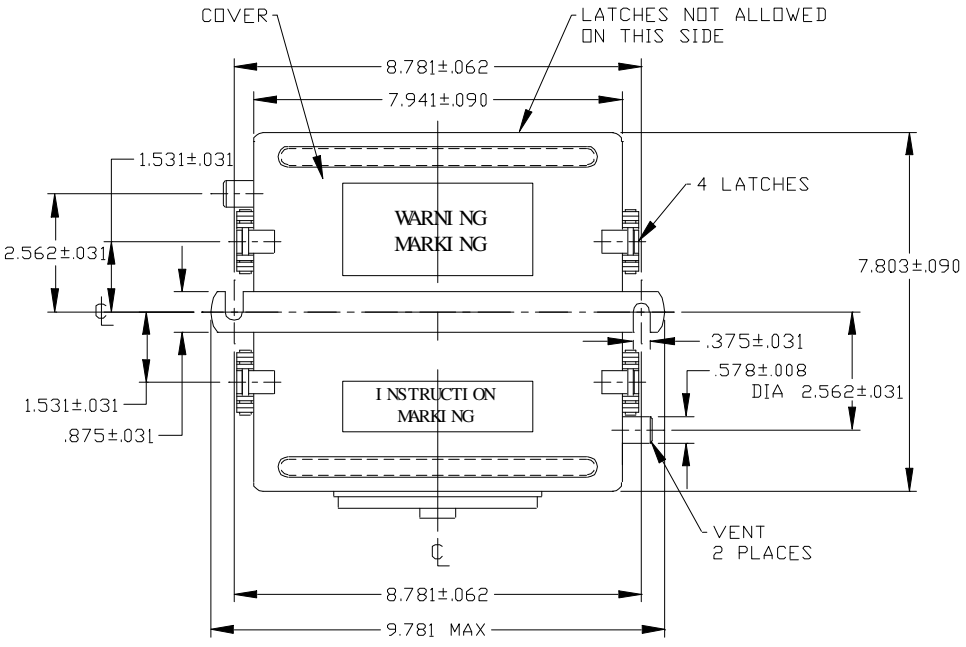


FIGURE 3: BATTERY ASSEMBLY BB-432B/A TOP VIEW

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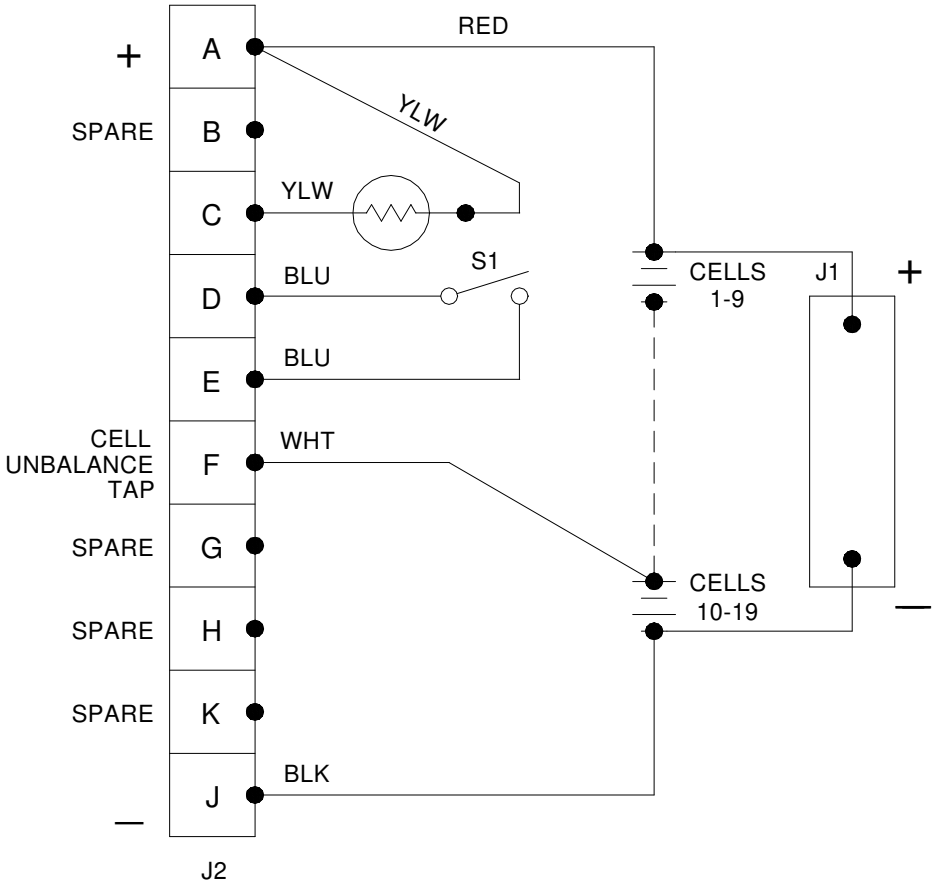


Figure 4 Battery, BB-432B/A, wiring harness.

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